

Commerce XI Maths App

Activities 4.1–4.3

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 4
Bivariate Frequency Distribution and Chi Square Statistic
Total questions: 18

Question 1 - MCQ · PG-52 · A4.1

A movie-preference survey has observed frequencies 25, 75, 65 and 35. What is the grand total?

- A. 200
- B. 100
- C. 180
- D. 400

Correct answer: 200

Explanation and steps

$25 + 75 + 65 + 35 = 200$.

Question 2 - MCQ · PG-52 · A4.1

	Melodramatic	Action	Total
Boys	25	75	100
Girls	65	35	100
Total	90	110	200

Find the row totals for boys and girls.

- A. 100 and 100
- B. 90 and 110
- C. 65 and 35
- D. 25 and 75

Correct answer: 100 and 100

Explanation and steps

Each row totals 100.

Question 3 - MCQ · PG-52 · A4.1

	Melodramatic	Action	Total
Boys	25	75	100
Girls	65	35	100
Total	90	110	200

Find the column totals for melodramatic and action movies.

- A. 90 and 110
- B. 100 and 100
- C. 65 and 35
- D. 25 and 75

Correct answer: 90 and 110

Explanation and steps

$25 + 65 = 90$ and $75 + 35 = 110$.

Question 4 - MCQ · PG-52 · A4.1

	Melodramatic	Action	Total
Boys	25	75	100
Girls	65	35	100
Total	90	110	200

Find the four expected frequencies in row order.

- A. 45, 55, 45, 55
- B. 25, 75, 65, 35
- C. 50, 50, 50, 50
- D. 90, 110, 90, 110

Correct answer: 45, 55, 45, 55

Explanation and steps

Use row total $\times$ column total / 200.

Question 5 - ENTER ANSWER · PG-52 · A4.1

	Melodramatic	Action
Boys	25	75
Girls	65	35

Calculate χ^2 statistic.

Correct answer: 32.32

Explanation and steps

The four contributions from expected counts 45 and 55 sum to approximately 32.32.

Question 6 - MCQ · PG-52 · A4.1

	Mathematics	Secretarial Practice	Total
Boys	50	25	75
Girls	39	36	75
Total	89	61	150

Find the expected frequency for boys choosing Mathematics.

- A. 44.5
- B. 30.5
- C. 39
- D. 50

Correct answer: 44.5

Explanation and steps

$$75 \times 89 / 150 = 44.5.$$

Question 7 - MCQ · PG-52 · A4.1

	Mathematics	Secretarial Practice	Total
Boys	50	25	75
Girls	39	36	75
Total	89	61	150

Find the expected frequency for boys choosing Secretarial Practice.

- A. 30.5
- B. 44.5
- C. 36
- D. 25

Correct answer: 30.5

Explanation and steps

$$75 \times 61 / 150 = 30.5.$$

Question 8 - ENTER ANSWER · PG-52 · A4.1

	Mathematics	Secretarial Practice
Boys	50	25
Girls	39	36

Calculate χ^2 statistic correct to two decimal places.

Correct answer: 3.86

Explanation and steps

Expected counts are 44.5 and 30.5 in each row; the contributions sum to approximately 3.86.

Question 9 - MCQ · PG-55 · A4.2

Students measure the height and weight of at least 20 students in their class and prepare a bivariate frequency table from the collected pairs.

Which measurements are collected?

- A. height and weight of at least 20 students
- B. age and income only
- C. marks in one subject only
- D. temperature only

Correct answer: height and weight of at least 20 students

Explanation and steps

The activity constructs a bivariate table from height and weight.

Question 10 - MCQ · PG-55 · A4.2

Students measure the height and weight of at least 20 students in their class and prepare a bivariate frequency table from the collected pairs.

The completed table

- A. depends on the observations collected
- B. is always identical
- C. must have zero frequencies
- D. has a fixed χ^2 value

Correct answer: depends on the observations collected

Explanation and steps

Different samples give different frequencies.

Question 11 - MCQ · PG-55 · A4.2

Students measure the height and weight of at least 20 students in their class and prepare a bivariate frequency table from the collected pairs.

Which are the two variables?

- A. height and weight
- B. gender and food preference
- C. age and blood group
- D. price and demand

Correct answer: height and weight

Explanation and steps

Both are measured for each student.

Question 12 - MCQ · PG-55 · A4.2

Students measure the height and weight of at least 20 students in their class and prepare a bivariate frequency table from the collected pairs.

Which is the suitable summary?

- A. bivariate frequency table
- B. single number only
- C. pie chart of one variable only
- D. list without pairs

Correct answer: bivariate frequency table

Explanation and steps

The table groups height and weight jointly.

Question 13 - MCQ · PG-55 · A4.3

A bivariate dataset contains height-weight measurements for 20 students.
What is the grand total?

- A. 20
- B. 6
- C. 7
- D. 72

Correct answer: 20

Explanation and steps

There are 20 student pairs.

Question 14 - MCQ · PG-55 · A4.3

Height classes are 115–125, 125–135, 135–145, 145–155, 155–165 and 165–175 cm.
What is the width of each height class interval?

- A. 10 cm
- B. 5 cm
- C. 15 cm
- D. 20 cm

Correct answer: 10 cm

Explanation and steps

The successive height class limits differ by 10 cm.

Question 15 - MCQ · PG-55 · A4.3

Weight classes are 62–64, 64–66, 66–68, 68–70 and 70–72 kg.
What is the width of each weight class interval?

- A. 2 kg
- B. 5 kg
- C. 10 kg
- D. 1 kg

Correct answer: 2 kg

Explanation and steps

The successive weight class limits differ by 2 kg.

Question 16 - MCQ · PG-55 · A4.3

A bivariate frequency table places height classes in columns and writes f_x below those columns. What does f_x represent?

- A. marginal frequencies of height
- B. conditional frequencies only
- C. expected frequencies
- D. chi-square contributions

Correct answer: marginal frequencies of height

Explanation and steps

Column totals summarize X.

Question 17 - MCQ · PG-55 · A4.3

A bivariate frequency table places weight classes in rows and writes f_y to the right of those rows. What does f_y represent?

- A. marginal frequencies of weight
- B. class marks
- C. expected frequencies
- D. chi-square contributions

Correct answer: marginal frequencies of weight

Explanation and steps

Row totals summarize Y.

Question 18 - MCQ · PG-55 · A4.3

Twenty height-weight pairs must be classified in a bivariate frequency table. What is required to complete the table?

- A. tallying each height-weight pair in its correct cell
- B. changing the original observations
- C. using only row headings
- D. ignoring marginal totals

Correct answer: tallying each height-weight pair in its correct cell

Explanation and steps

Every ordered pair is classified once.